1.) We'll use the Cauchy criterion. Fix $\varepsilon > 0$.
As $\{b_n\}$ is bounded, $\exists M > 0$ s.t.

$$|b_n| \leq M \quad \forall n \geq 1.$$

As $\sum |a_n|$ converges, $\exists N \in \mathbb{N}$ s.t.

$$\sum_{k=m}^{n} |a_k| < \frac{\varepsilon}{M} \quad \forall n \geq m \geq N.$$

So, for $n \geq m \geq N$ we have

$$\left|\sum_{k=m}^{n} a_k b_k\right| \leq \sum_{k=m}^{n} |a_k| \cdot |b_k|$$

$$\leq M \sum_{k=m}^{n} |a_k| < M \cdot \frac{\varepsilon}{M} = \varepsilon.$$

Therefore $\sum a_n b_n$ converges by the Cauchy criterion. $\square$

2.) (a.) Set $a_n = \frac{1}{\sqrt{n^n}}$. Then

$$|a_n|^{\frac{1}{n}} = \frac{1}{(n^{\frac{n}{2}})^{\frac{1}{n}}} = \frac{1}{n^{\frac{1}{2}}} = \sqrt{\frac{1}{n}} \to 0 \text{ as } n \to \infty,$$

and so $\limsup |a_n|^{\frac{1}{n}} = 0 < 1$. Therefore the series $\boxed{\text{converges}}$ by the root test.

(b.) Set $s_n = \sum_{k=1}^{n} \left(\frac{1}{\sqrt{k}} - \frac{1}{\sqrt{k+1}} \right)$.

Claim: $s_n = 1 - \frac{1}{\sqrt{n+1}} \quad \forall n \geq 1$. We'll argue by induction.

- Base case: For $n = 1$, we have $s_1 = \frac{1}{\sqrt{1}} - \frac{1}{\sqrt{2}}$
$= 1 - \frac{1}{\sqrt{2}}$.

- Inductive step: Suppose $s_n = 1 - \frac{1}{\sqrt{n+1}}$ for some $n \geq 1$. Then

$$s_{n+1} = s_n + \left(\frac{1}{\sqrt{n+1}} - \frac{1}{\sqrt{n+2}}\right) = 1 - \frac{1}{\sqrt{n+1}} + \frac{1}{\sqrt{n+1}} - \frac{1}{\sqrt{n+2}}$$
$$= 1 - \frac{1}{\sqrt{n+2}}$$

So the formula holds for $n+1$.

Together, we conclude $s_n = 1 - \frac{1}{\sqrt{n+1}}$ $\forall n \geq 1$ by induction.

We know $\frac{1}{n} \to 0$ as $n \to \infty$ from lecture, and so the subsequence $\frac{1}{n+1} \to 0$ as well. So, by the limit laws,

$$s_n = 1 - \frac{1}{\sqrt{n+1}} = 1 - \sqrt{\frac{1}{n+1}} \to 1 \quad \text{as } n \to \infty.$$

So the series $\boxed{\text{converges}}$ to 1. $\square$

3.) (a.) <u>Disprove</u>: Consider $d(x,y) = |x-y|$, $\tilde{d}(x,y) = \min\{d(x,y), 1\}$, and $r = 2$. Then

$$\{x \in \mathbb{R} : \tilde{d}(x,0) < 2\} = \mathbb{R}$$
$$\{x \in \mathbb{R} : d(x,0) < 2\} = (-2,2)$$

and $\mathbb{R} \not\subseteq (-2,2)$.

(b.) <u>Prove</u>: Fix $r > 0$. Then

$$y \in \{x \in \mathbb{R} : d(x,0) < r\} \Rightarrow d(y,0) < r$$
$$\Rightarrow \tilde{d}(y,0) \leq d(y,0) < r$$

$$\Rightarrow y \in \{x \in \mathbb{R} : \tilde{d}(x,0) < r\}. \qquad \square$$

4.) $\underline{\Rightarrow|}$: Suppose $x^n \to x$ in $(\mathbb{R}^2, d_1)$. Then $d_1(x^n, x) \to 0$ as $n \to \infty$. Note that

$$0 \leq |x_1^n - x_1| \leq |x_1^n - x_1| + |x_2^n - x_2| = d_1(x^n, x) \quad \forall n \geq 1.$$

So $\lim_{n\to\infty} x_1^n = x_1$ by the squeeze theorem. Similarly,

$$0 \leq |x_2^n - x_2| \leq |x_1^n - x_1| + |x_2^n - x_2| = d_1(x^n, x),$$

and so $\lim_{n\to\infty} x_2^n = x_2$.

$\underline{\Leftarrow|}$: Suppose $\lim_{n\to\infty} x_1^n = x_1$ and $\lim_{n\to\infty} x_2^n = x_2$. Then $|x_1^n - x_1| \to 0$ and $|x_2^n - x_2| \to 0$ as $n \to \infty$, and so by the limit law for addition,

$$|x_1^n - x_1| + |x_2^n - x_2| \to 0 \quad \text{as } n \to \infty.$$

Therefore $d_1(x^n, x) \to 0$ as $n \to \infty$, and so $x^n \to x$ in $(\mathbb{R}^2, d_1)$. $\square$